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Semiconductor device

Abstract

A semiconductor device includes a semiconductor layer, a first conductor disposed on the semiconductor layer, a second conductor disposed on the semiconductor layer so as to be separated from the first conductor, a relay portion that is formed on the semiconductor layer so as to straddle the first conductor and the second conductor and that is made of a semiconductor having a first conductivity type region and a second conductivity type region, a first contact by which the first conductivity type region and the second conductivity type region are electrically connected to the first conductor, and a second contact that electrically connects the first conductivity type region of the relay portion and the second conductor together and that is insulated from the second conductivity type region.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a semiconductor device.

BACKGROUND ART

(2) Patent Literature 1 discloses a semiconductor device which includes a semiconductor substrate, a first conductivity type drift layer formed at the semiconductor substrate, a second conductivity type body region and a first conductivity type source region that are formed at the drift layer, a gate insulating film disposed on a part of the body region sandwiched between the drift layer and the source region, a gate electrode disposed so as to face the part of the body region sandwiched between the drift layer and the source region with the gate insulating film between the gate electrode and the part of the body region, a source electrode disposed on the semiconductor substrate, and a gate pad that is formed on the semiconductor substrate and that is electrically connected to the gate electrode through a gate wiring.

CITATION LIST

Patent Literature

(3) Patent Literature 1: Japanese Patent Application Publication No. 2017-143188

SUMMARY OF INVENTION

Technical Problem

(4) In controlling the voltage of the semiconductor device, the amount of noise caused by ringing when the voltage is turned on and the amount of noise caused by ringing when the voltage is turned off are never completely equal to each other, and there is a case in which only the noise caused when the voltage is turned on is intended to be reduced or only the noise caused when the voltage is turned off is intended to be reduced.

(5) For example, referring to Patent Literature 1, when a voltage is applied from the gate pad to the gate electrode, ringing easily occurs when the voltage is turned on, whereas ringing does not easily occur when the voltage is turned off, and therefore it is preferable to reduce only the noise produced when the voltage is turned on.

(6) Therefore, it is considered that a circuit in which a pair of resistors connected in parallel with each other and in which a diode is connected to only one of the resistors in series is disposed outside the semiconductor device. Hence, when a forward current of the diode flows to the parallel circuit, the current flows through both of a pair of current paths, and therefore it is possible to make resistance smaller, whereas when a reverse current of the diode flows thereto, the current flows through only one (to which the diode is not connected) of the pair of current paths, and therefore it is possible to make resistance larger. Therefore, it is expected that the aforementioned problem of reducing only the noise caused when the voltage is turned on or only the noise caused when the voltage is turned off will be solved by selectively increasing resistance when ringing easily occurs.

(7) However, at least one chip is required besides the semiconductor device, and the space efficiency is forced to be reduced when the semiconductor device is mounted.

(8) An object of the present invention is to provide a semiconductor device that is capable of performing control so that resistances become different from each other between a case in which an electric current flows in a direction from a first conductor toward a second conductor and a case in which an electric current flows in a direction opposite thereto while maintaining the space efficiency when the semiconductor device is mounted.

(9) Another object of the present invention is to provide a semiconductor device that is capable of appropriately controlling the behavior of a gate current when a functional element is turned on/off while maintaining the space efficiency when the semiconductor device is mounted.

Solution to Problem

(10) A semiconductor device according to one preferred embodiment of the present invention includes a semiconductor layer, a first conductor disposed on the semiconductor layer, a second conductor disposed on the semiconductor layer so as to be separated from the first conductor, a relay portion that is formed on the semiconductor layer so as to straddle the first conductor and the second conductor and that is made of a semiconductor having a first conductivity type region and a second conductivity type region, a first contact by which the first conductivity type region and the second conductivity type region are electrically connected to the first conductor, and a second contact that electrically connects the first conductivity type region of the relay portion and the second conductor together and that is insulated from the second conductivity type region.

(11) For example, if the first conductivity type is a p type and if the second conductivity type is an n type, the first conductor is connected to both the p type region and the n type region through the first contact, and the second conductor is connected to only the p type region through the second contact.

(12) When a positive voltage with respect to the second conductor is applied to the first conductor, the flow of an electric current between the first conductor and the second conductor takes a direction from the first conductor toward the second conductor. In this case, a reverse current will flow to a pn junction between the first conductivity type region (p type region) and the second conductor type region (n type region). Therefore, the current path is limited to the path of (1) the first conductor.fwdarw.the first contact.fwdarw.the first conductivity type region (p type region).fwdarw.the second contact.fwdarw.the second conductor, and an electric current does not flow or hardly flows to the path of (2) the first conductor.fwdarw.the first contact.fwdarw.the second conductivity type region (n type region).fwdarw.the pn junction.fwdarw.the first conductivity type region (p type region).fwdarw.the second contact.fwdarw.the second conductor.

(13) On the other hand, when a positive voltage with respect to the first conductor is applied to the second conductor, the flow of an electric current between the first conductor and the second conductor takes a direction from the second conductor toward the first conductor. In this case, a forward current will flow to the pn junction between the first conductivity type region (p type region) and the second conductor type region (n type region). Therefore, it is possible to use two paths in total as current paths, i.e., it is possible to use the path of (3) the second conductor.fwdarw.the second contact.fwdarw.the first conductivity type region (p type region).fwdarw.the first contact.fwdarw.the first conductor and the path of (4) the second conductor.fwdarw.the second contact.fwdarw.the first conductivity type region (p type region).fwdarw.the pn junction.fwdarw.the second conductivity type region (n type region) the first contact.fwdarw.the first conductor.

(14) In other words, in the former case, the number of current paths is one, hence making it possible to relatively heighten resistance, and in the latter case, the number of current paths is two, hence making it possible to relatively make resistance lower than that in the former case. If the first conductivity type is an n type and if the second conductivity type is a p type, the number of current paths is two when a positive voltage with respect to the second conductor is applied to the first conductor, and the number of current paths is one when a positive voltage with respect to the first conductor is applied to the second conductor.

(15) As thus described, the number of current paths can be changed according to the positive/negative direction of a voltage, and therefore it is possible to make resistances different from each other between a case in which an electric current flows in a direction from the first conductor toward the second conductor and a case in which an electric current flows in a direction opposite thereto. Moreover, it is possible to perform such current control inside the semiconductor device, and therefore it is also possible to maintain the space efficiency when the semiconductor device is mounted.

(16) The semiconductor device according to one preferred embodiment of the present invention may further include a functional element formed at the semiconductor layer, and, in the semiconductor device, the first conductor may include an external terminal to which electric power is supplied from outside, and the second conductor may include a wiring that supplies electric power supplied to the first conductor to the functional element.

(17) In the semiconductor device according to one preferred embodiment of the present invention, the functional element may be an element including a gate electrode that controls an electric current that flows to the functional element, and the external terminal may include a gate pad to which an electroconductive bonding member is bonded from outside, and the wiring may include a gate wiring that supplies electric power supplied to the gate pad to the gate electrode, and the first conductivity type region may be a p type region, and the second conductivity type region may be an n type region.

(18) According to this arrangement, the number of paths of gate current that flows when the functional element is turned on differs from the number of paths of gate current that flows when the functional element is turned off, and resistance when the functional element is turned on differs from resistance when the functional element is turned off. Therefore, it is possible to appropriately control the behavior of a gate current when the functional element is turned on/off.

(19) In the semiconductor device according to one preferred embodiment of the present invention, the gate wiring may include a gate finger disposed at an outer peripheral portion of the semiconductor layer so as to surround the functional element.

(20) In the semiconductor device according to one preferred embodiment of the present invention, the relay portion may be disposed closer to the semiconductor layer than the gate pad and the gate wiring, and the first conductivity type region and the second conductivity type region may each extend from a region below the gate pad to a region below the gate wiring so that a boundary portion between the first conductivity type region and the second conductivity type region intersects the gate pad and the gate wiring.

(21) In the semiconductor device according to one preferred embodiment of the present invention, the functional element may include a field-effect transistor that has a body region selectively formed at a surface portion of the semiconductor layer, a source region formed at an inner portion of the body region, and the gate electrode facing a part of the body region through a gate insulating film.

(22) In the semiconductor device according to one preferred embodiment of the present invention, the body region may include a plurality of body regions that extend in a striped manner with intervals from each other.

(23) In the semiconductor device according to one preferred embodiment of the present invention, the relay portion may include a second conductivity type layer as the second conductivity type region and the first conductivity type region selectively formed at a surface portion of the second conductivity type layer.

(24) In the semiconductor device according to one preferred embodiment of the present invention, a thickness of the second conductivity type layer may be 0.1 μm to 10 μm , and a depth of the first conductivity type region from a surface of the second conductivity type layer may be 0.1 μm to 10 μm .

(25) In the semiconductor device according to one preferred embodiment of the present invention,

the relay portion may include a second conductivity type layer as the second conductivity type region and a first conductivity type layer as the first conductivity type region, the first conductivity type layer adjoining the second conductivity type layer and being contiguous to the second conductivity type layer.

(26) The semiconductor device according to one preferred embodiment of the present invention may further include a slit that is formed on an extension line of a boundary portion between the second conductivity type layer and the first conductivity type layer and by which the second conductivity type layer and the first conductivity type layer are partially separated from each other.

(27) In the semiconductor device according to one preferred embodiment of the present invention, the first contact may be formed so as to straddle the first conductivity type region and the second conductivity type region of the relay portion.

(28) In the semiconductor device according to one preferred embodiment of the present invention, the first contact may include a one-side first contact that is connected to the first conductivity type region of the relay portion and an opposite-side first contact that is apart from the one-side first contact and that is connected to the second conductivity type region of the relay portion.

(29) In the semiconductor device according to one preferred embodiment of the present invention, an impurity concentration of the first conductivity type region may be $1.0 \times 10^{19} \text{ cm}^{-3}$ to $1.0 \times 10^{21} \text{ cm}^{-3}$, and an impurity concentration of the second conductivity type region may be $1.0 \times 10^{19} \text{ cm}^{-3}$ to $1.0 \times 10^{21} \text{ cm}^{-3}$.

(30) In the semiconductor device according to one preferred embodiment of the present invention, the first conductor and the second conductor may each be made of aluminum, and the relay portion may be made of polysilicon.

(31) In the semiconductor device according to one preferred embodiment of the present invention, the semiconductor layer may include a silicon substrate.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic plan view of a semiconductor device according to a preferred embodiment of the present invention.

(2) FIG. 2 is a schematic cross-sectional view of the semiconductor device.

(3) FIG. 3 is an enlarged view of a region surrounded by the alternate long and two short dashes line III of FIG. 1.

(4) FIG. 4 is a schematic cross-sectional perspective view showing a structure of a portion below a gate pad.

(5) FIG. 5 is a cross-sectional view showing a cross section taken along line V-V of FIG. 3.

(6) FIG. 6 is a cross-sectional view showing a cross section taken along line VI-VI of FIG. 3.

(7) FIG. 7 is a flowchart showing a part of a process of manufacturing the semiconductor device.

(8) FIG. 8 is a view showing an equivalent circuit (when turned on) of a gate electrode of the semiconductor device.

(9) FIG. 9 is a view showing the equivalent circuit (when turned off) of the gate electrode of the semiconductor device.

(10) FIG. 10 is a diagram showing I-V characteristics (when turned on) of the gate electrode of the semiconductor device.

(11) FIG. 11 is diagram showing I-V characteristics (when turned off) of the gate electrode of the semiconductor device.

(12) FIG. 12 is a view showing a modification of the semiconductor device.

(13) FIG. 13 is a view showing a modification of the semiconductor device.

(14) FIG. 14 is a view showing a modification of the semiconductor device.

(15) FIG. 15 is a measurement circuit diagram used in examples.

(16) FIG. 16 is a schematic plan view showing a structure of a portion below a gate pad of Example 1.

(17) FIG. 17 is a schematic plan view showing a structure of a portion below a gate pad of Example 2.

(18) FIG. 18 is a schematic plan view showing a structure of a portion below a gate pad of Example 3.

(19) FIG. 19 is a schematic plan view showing a structure of a portion below a gate pad of Example 4.

(20) FIG. 20 is a schematic plan view showing a structure of a portion below a gate pad of Example 5.

(21) FIG. 21 is a schematic plan view showing a structure of a portion below a gate pad of Example 6.

(22) FIG. 22 is a schematic plan view showing a structure of a portion below a gate pad of Example 7.

(23) FIG. 23 is a view showing a time-dependent change of an electric current when the gate voltage is turned on/off with respect to Examples 1 to 3.

(24) FIG. 24 is a view showing a time-dependent change of an electric current when the gate voltage is turned on/off with respect to Examples 2, 4, and 5.

(25) FIG. 25 is a view showing a time-dependent change of an electric current when the gate voltage is turned on/off with respect to Examples 2, 6, and 7.

(26) FIG. 26 is a schematic plan view showing a structure of a portion below a gate pad of Example 8.

(27) FIG. 27 is a schematic plan view showing a structure of a portion below a gate pad of Example 9.

DESCRIPTION OF EMBODIMENTS

(28) Preferred embodiments of the present invention will be hereinafter described in detail with reference to the accompanying drawings.

(29) FIG. 1 is a schematic plan view of a semiconductor device 1 according to a preferred embodiment of the present invention. For clarity, electrode films 5 and 6 are hatched as shown in FIG. 1.

(30) The semiconductor device 1 includes a semiconductor substrate 2 that is an example of a semiconductor layer of the present invention and that is formed in a quadrangular shape in a plan view. A length L1 (i.e., length along lateral surfaces 2A and 2C of the semiconductor substrate 2 in FIG. 1) in a first direction of the semiconductor substrate 2 may be, for example, 1.0 mm to 9.0 mm, and a length L2 (i.e., length along lateral surfaces 2B and 2D of the semiconductor substrate 2 in FIG. 1) in a second direction perpendicular to the first direction may be, for example, 1.0 mm to 9.0 mm.

(31) The semiconductor substrate 2 includes an active portion 3 in its central region in a plan view. The active portion 3 is a region in which a unit cell 19 described later is chiefly formed, and is a region in which an electric current flows in a thickness direction of the semiconductor substrate 2 when a source-to-drain space of the semiconductor device 1 is in an electrically conductive state (i.e., when turned on). The semiconductor substrate 2 additionally includes an outer peripheral portion 4 around the active portion 3.

(32) The semiconductor device 1 includes a source electrode film 5 and a gate electrode film 6. These electrode films 5 and 6 are formed so as to be separated from each other by patterning of a common electrode film.

(33) The source electrode film 5 is formed in a substantially quadrangular shape in a plan view with which most of the active portion 3 is covered. A concave portion 7 that is concaved toward an inward side of the source electrode film 5 is formed at one lateral portion of the source electrode

film 5 (i.e., lateral portion along the lateral surface 2C of the semiconductor substrate 2 in FIG. 1). The concave portion 7 is provided to effectively secure an arrangement space for a first conductive film 9 described later. The source electrode film 5 is selectively covered with a surface insulating film 31 (see FIG. 5 and FIG. 6), and a part of the source electrode film 5 is exposed as a source pad 8. A joining member, such as a bonding wire, is connected to the source pad 8.

(34) The gate electrode film 6 includes the first conductive film 9 that is an example of a first conductor of the present invention and a second conductive film 10 that is an example of a second conductor of the present invention.

(35) The first conductive film 9 includes a part, which is selectively exposed from the surface insulating film 31, of the gate electrode film 6 covered with the surface insulating film 31 (see FIG. 5 and FIG. 6). A joining member, such as a bonding wire, is connected to the first conductive film 9. In other words, the first conductive film 9 functions as an external terminal on the gate side in the semiconductor device 1. The first conductive film 9 is disposed in an inward region of the concave portion 7 of the source electrode film 5 in a plan view.

(36) The second conductive film 10 is formed in a linear shape along the lateral surfaces 2A to 2D of the semiconductor substrate 2 from the first conductive film 9. In the present preferred embodiment, the second conductive film 10 is formed in a closed annular shape that surrounds the source electrode film 5. The second conductive film 10 functions as a gate wiring (gate finger) that supplies electric power supplied to the first conductive film 9 to a gate electrode 17 described later. The second conductive film 10 is not necessarily required to have the closed annular shape, and may be formed in a partially-opened shape. The second conductive film 10 may be formed in, for example, a shape that is opened on the side opposite to the first conductive film 9. Additionally, the second conductive film 10 is covered with the surface insulating film 31 (see FIG. 5 and FIG. 6).

(37) FIG. 2 is a schematic cross-sectional view of the semiconductor device 1. FIG. 3 is an enlarged view of a region surrounded by the alternate long and two short dashes line III of FIG. 1. FIG. 4 is a schematic cross-sectional perspective view showing a structure of a portion below the first conductive film 9. FIG. 5 is a cross-sectional view showing a cross section taken along line V-V of FIG. 3. FIG. 6 is a cross-sectional view showing a cross section taken along line VI-VI of FIG. 3. In FIG. 2, a configuration formed on an interlayer insulating film 25 is omitted. In FIG. 5, with respect to a relay portion 21, only a p type region 23 is shown for convenience, and an n type layer 22 below the p type region 23 is omitted.

(38) In the present preferred embodiment, the semiconductor device 1 is an n-channel type MISFET (Metal Insulator Semiconductor Field Effect Transistor) that is an example of a functional element of the present invention.

(39) The semiconductor device 1 includes an n⁺ type drain layer 11, an n⁻ type base layer 12, a p type body region 13, an n⁺ type source region 14, a p⁺ type body contact region 15, a gate insulating film 16, a gate electrode 17, and a drain electrode 18. The semiconductor substrate 2 of FIG. 1 may be a concept for which the n⁺ type drain layer 11 and the n⁻ type base layer 12 are combined together.

(40) The n⁺ type drain layer 11 may be made of an n⁺ type semiconductor substrate (for example, silicon substrate). Besides, the n⁺ type drain layer 11 may be a substrate, such as an SiC substrate or a GaN substrate, that is generally employed in transistors. The n⁺ type semiconductor substrate may be a semiconductor substrate that has undergone crystal growth while being doped with n type impurities. P (phosphorus), As (arsenic), Sb (antimony), etc., can be applied as the n type impurities. The impurity concentration of the n⁺ type drain layer 11 is, for example, about $1.0 \times 10^{18} \text{ cm}^{-3}$ to $5.0 \times 10^{20} \text{ cm}^{-3}$. The thickness of the n⁺ type drain layer 11 is, for example, 1 μm to 5 μm .

(41) The n⁻ type base layer 12 is a semiconductor layer into which n type impurities are implanted. More specifically, the n⁻ type base layer 12 may be an n type epitaxial layer that has been epitaxially grown while being implanted with n type impurities on the n⁺ type drain layer 11. The

aforementioned ones can be applied as the n type impurities. The impurity concentration of the n- type base layer **12** is lower than that of the n+ type drain layer **11**, and is, for example, about $1.0 \times 10^{10} \text{ cm}^{-3}$ to $1.0 \times 10^{16} \text{ cm}^{-3}$. The thickness of the n- type base layer **12** is, for example, 10 μm to 50 μm .

(42) The p type body region **13** is a semiconductor layer into which p type impurities are implanted. More specifically, that may be a semiconductor layer formed by performing ion implantation of p type impurities into a surface of the n- type base layer **12**. B (boron), Al (aluminum), Ga (gallium), etc., can be applied as the p type impurities. The impurity concentration of the p type body region **13** is, for example, about $1.0 \times 10^{15} \text{ cm}^{-3}$ to $1.0 \times 10^{19} \text{ cm}^{-3}$.

(43) The p type body region **13** is selectively formed at a surface portion of the n- type base layer **12**. In the present preferred embodiment, a plurality of p type body regions **13** are formed parallel to each other in a striped manner as shown in FIG. 2, and, for example, may extend in a direction along the lateral surfaces **2A** and **2C** of the semiconductor substrate **2** (see FIG. 1). The plurality of p type body regions **13** may be arranged in a matrix manner in the surface portion of the n- type base layer **12**. The width of each of the p type body regions **13** is, for example, 3 μm to 10 μm . A region including each of the p type body regions **13** and the n- type base layer **12** therearound constitutes a unit cell **19**. In other words, the semiconductor device **1** has many (a plurality of) unit cells **19** arranged in a striped manner in a plan view in the layout of FIG. 2. Additionally, in FIG. 2, the width (cell pitch) of the adjoining unit cells **19** is, for example, 5 μm to 20 μm .

(44) The n+ type source region **14** is formed in an inward region of the p type body region **13** of each of the unit cells **19**. In this region, the n+ type source region **14** is selectively formed at a surface portion of the p type body region **13**. The n+ type source region **14** may be formed by selectively performing ion implantation of n type impurities into the p type body region **13**. Examples of the n type impurities are as mentioned above. The impurity concentration of the n+ type source region **14** is higher than that of the n- type base layer **12**, and is, for example, about $1.0 \times 10^{18} \text{ cm}^{-3}$ to $5.0 \times 10^{20} \text{ cm}^{-3}$.

(45) The n+ type source region **14** is formed inside the p type body region **13** so as to be placed at an inward position by a predetermined distance from a circumferential edge of the p type body region **13** (i.e., from an interface between the p type body region **13** and the n- type base layer **12**). Hence, in a surface layer region of the semiconductor layer including the n- type base layer **12** and the p type body region **13**, etc., the surface portion of the p type body region **13** is interposed between the n+ type source region **14** and the n- type base layer **12**, and the surface portion interposed therebetween provides a channel region **20**.

(46) In the present preferred embodiment, the n+ type source region **14** is formed in a striped manner. The channel region **20** has a stripe shape in accordance with the shape of the n+ type source region **14**.

(47) The p+ type body contact region **15** is selectively formed at the surface portion of the p type body region **13**. The p+ type body contact region **15** may be formed by selectively performing ion implantation of p type impurities into the p type body region **13**. Examples of the p type impurities are as mentioned above. The impurity concentration of the p+ type body contact region **15** is higher than that of the p type body region **13**, and is, for example, about $5.0 \times 10^{17} \text{ cm}^{-3}$ to $1.0 \times 10^{19} \text{ cm}^{-3}$.

(48) The p+ type body contact region **15** passes through the n+ type source region **14**, and extends toward the n+ type drain layer **11** up to a halfway position of the p type body region **13**.

(49) The gate insulating film **16** may be made of, for example, a silicon oxide film, a silicon nitride film, a silicon oxynitride film, a hafnium oxide film, an alumina film, a tantalum oxide film, or the like. Referring to FIG. 2, the gate insulating film **16** is formed so as to cover at least a surface of the p type body region **13** in the channel region **20**. In the present preferred embodiment, the gate insulating film **16** is formed so as to cover a part of the n+ type source region **14**, the channel

region **20**, and the surface of the n⁻ type base layer **12**. More clearly, the gate insulating film **16** is formed with a pattern that has an opening in the p⁺ type body contact region **15** of each of the unit cells **19** and an inner edge region of the n⁺ type source region **14** continuous with the p⁺ type body contact region **15**. Additionally, referring to FIG. 5 and FIG. 6, the gate insulating film **16** is also formed at a portion below the first conductive film **9**.

(50) The gate electrode **17** is formed so as to face the channel region **20** with the gate insulating film **16** between the gate electrode **17** and the channel region **20**. The gate electrode **17** may be made of, for example, polysilicon whose resistance has been lowered by injecting impurities.

(51) Referring to FIG. 2, in the active portion **3**, the gate electrode **17** is formed in substantially the same pattern as that of the gate insulating film **16**, and covers a surface of the gate insulating film **16**. In other words, the gate electrode **17** is disposed above a part of the n⁺ type source region **14**, the channel region **20**, and the surface of the n⁻ type base layer **12**. More clearly, the gate electrode **17** is formed with a pattern that has an opening in the p⁺ type body contact region **15** of each of the unit cells **19** and an inner edge region of the n⁺ type source region **14** continuous with the p⁺ type body contact region **15**. In other words, the gate electrode **17** is formed so as to control the unit cells **19** in common. Hence, a planar gate structure is formed.

(52) On the other hand, referring to FIG. 3 to FIG. 6, the relay portion **21** is formed on the gate insulating film **16** so as to face the first conductive film **9** and the second conductive film **10**. The relay portion **21** relays an electrical connection between the first conductive film **9** and the second conductive film **10**.

(53) Referring to FIG. 3, the relay portion **21** is formed so as to straddle the first conductive film **9** and the second conductive film **10** in a portion below the first and second conductive films **9** and **10** (more specifically, between the n⁻ type base layer **12** and each of the first and second conductive films **9** and **10**).

(54) Referring to FIG. 4, the relay portion **21** includes an n type layer **22** that is formed in a quadrangular shape in a plan view and that is an example of a second conductivity type region and an example of a second conductivity type layer of the present invention and a p type region **23** that is selectively formed at a surface portion of the n type layer **22** and that is an example of a first conductivity type region of the present invention. Hence, in the relay portion **21**, a diode **34** is formed at a boundary portion **24** between the n type layer **22** and the p type region **23** by means of pn junction between the n type layer **22** and the p type region **23**.

(55) In the present preferred embodiment, the p type region **23** is formed from one end portion of the n type layer **22** in a direction perpendicular to the lateral surface **2C** of the semiconductor substrate **2** in a plan view to the other end portion thereof so as to divide the n type layer **22** into an n type part and a p type part in the perpendicular direction. Hence, the n type layer **22** and the p type region **23** each extend from a region below the first conductive film **9** to a region below the second conductive film **10** so that the boundary portion **24** intersects the first conductive film **9** and the second conductive film **10**.

(56) In the present preferred embodiment, the relay portion **21** may be made of the same material (for example, polysilicon) as that of the gate electrode **17**. The thickness of the n type layer **22** is, for example, 0.1 μm to 10 μm . On the other hand, the depth of the p type region **23** from the surface of the n type layer **22** is, for example, 0.1 μm to 10 μm . Referring to FIG. 4, the p type region **23** is selectively formed at the surface portion of the n type layer **22**, and the n type part of the n type layer **22** comes around below the p type region **23**, and this n type part is disposed between the p type region **23** and the gate insulating film **16** in the present preferred embodiment. However, the range of the p type region **23** is not limited to this, and the p type region **23** may be formed over the entirety in the depth direction of the n type layer **22** from the surface of the n type layer **22** to the gate insulating film **16**, for example, as shown by the broken line **23'** in FIG. 4.

(57) The impurity concentration of the n type layer **22** is, for example, $1.0 \times 10^{19} \text{ cm}^{-3}$ to $1.0 \times 10^{21} \text{ cm}^{-3}$. The impurity concentration of the p type region **23** is, for example,

1.0×10^{sup.19} cm.^{sup.-3} to 1.0×10^{sup.21} cm.^{sup.-3}. Examples of both the n type impurities and the p type impurities are as mentioned above.

(58) An interlayer insulating film **25** is formed on the n- type base layer **12** so as to cover the gate electrode **17** and the relay portion **21**. The interlayer insulating film **25** is made of an insulating material, such as a silicon oxide film, a silicon nitride film, or TEOS (tetraethoxysilane).

(59) Referring to FIG. 2, a contact hole **26** by which the p+ type body contact region **15** and the n+ type source region **14** of each of the unit cells **19** are exposed is formed in the interlayer insulating film **25**. The contact hole **26** is formed so as to penetrate through the interlayer insulating film **25** and the gate insulating film **16**.

(60) Referring to FIG. 3, FIG. 5, and FIG. 6, a contact hole **27** by which an inner end portion of the relay portion **21** in the direction perpendicular to the lateral surface **2C** of the semiconductor substrate **2** is exposed and a contact hole **28** by which an outer end portion on the side opposite to the inner end portion is exposed are formed in the interlayer insulating film **25**. The contact hole **27** is formed so as to straddle the boundary portion **24** between the n type layer **22** and the p type region **23**, and exposes both the n type layer **22** and the p type region **23**. On the other hand, the contact hole **28** exposes only one of the n type layer **22** and the p type region **23** (in the present preferred embodiment, only the p type region **23**).

(61) The source electrode film **5** is made of aluminum or other metals. The source electrode film **5** is formed so as to selectively cover a surface of the interlayer insulating film **25** and so as to be embedded in the contact hole **26**. Hence, the source electrode film **5** is ohmically connected to the n+ type source region **14**. Therefore, the source electrode film **5** is connected to the unit cells **19** in parallel, and is arranged so that an entire current which flows to the unit cells **19** flows.

Additionally, the source electrode film **5** is ohmically connected to the p+ type body contact region **15** of each of the unit cells **19** through the contact hole **26**, and stabilizes the electric potential of the p type body region **13**.

(62) The gate electrode film **6** is made of aluminum or other metals. Referring to FIG. 3 to FIG. 6, the first conductive film **9** is formed so as to selectively cover the surface of the interlayer insulating film **25** and so as to be embedded in the contact hole **27**. Hence, the first conductive film **9** is electrically connected to both the n type layer **22** and the p type region **23**. In other words, the part of the first conductive film **9** which has been embedded in the contact hole **27** straddles both the n type layer **22** and the p type region **23** so as to serve as a first contact **29**, and is electrically connected to the n type layer **22** and the p type region **23**. On the other hand, referring to FIG. 3 to FIG. 6, the second conductive film **10** is formed so as to selectively cover the surface of the interlayer insulating film **25** and so as to be embedded in the contact hole **28**. Hence, the second conductive film **10** is electrically connected to only the p type region **23**, and is physically insulated from the n type layer **22**. In other words, the part of the second conductive film **10** which has been embedded in the contact hole **28** is electrically connected to only the p type region **23** so as to serve as a second contact **30**.

(63) Although the first contact **29** and the second contact **30** are formed by using the same materials integrally with the first conductive film **9** and the second conductive film **10**, respectively, in the present preferred embodiment, the first contact **29** and the second contact **30** may be made of different materials. For example, titanium, titanium nitride, tungsten, etc., can be used as other materials for the first contact **29** and the second contact **30**.

(64) The surface insulating film **31** is formed on a topmost surface of the semiconductor substrate **2** so as to cover the source electrode film **5** and the gate electrode film **6**. The surface insulating film **31** is made of an insulating material, such as a silicon nitride film or a polyimide film. Referring to FIG. 5 and FIG. 6, a pad opening **33** by which a part of the first conductive film **9** is exposed as a gate pad **32** is formed in the surface insulating film **31**. A pad opening (not shown) by which a part of the source electrode film **5** is exposed as the source pad **8** is formed in the surface insulating film **31**.

(65) The drain electrode **18** is made of aluminum or other metals. The drain electrode **18** is formed so as to come into contact with a rear surface of the n⁺ type drain layer **11**. Hence, the drain electrode **18** is connected to the unit cells **19** in parallel, and is arranged so that an entire current which flows to the unit cells **19** flows.

(66) Next, a method of manufacturing the semiconductor device **1** will be described with reference to FIG. 7.

(67) To manufacture the semiconductor device **1**, the n⁻ type base layer **12** is formed on the n⁺ type drain layer **11**, for example, by means of epitaxial growth (S1).

(68) Thereafter, p type ions are selectively implanted into the surface of the n⁻ type base layer **12**, and annealing treatment (1000° C. to 1200° C.) is performed, and, as a result, the p type body region **13** is formed (S2).

(69) Thereafter, n type ions are selectively implanted into the surface of the p type body region **13**, and annealing treatment (1000° C. to 1200° C.) is performed, and, as a result, the n⁺ type source region **14** is formed (S3).

(70) Thereafter, p type ions are selectively implanted into the surface of the p type body region **13**, and annealing treatment (1000° C. to 1200° C.) is performed, and, as a result, the p⁺ type body contact region **15** is formed (S4).

(71) Thereafter, the gate insulating film **16** is formed on the n⁻ type base layer **12** (S5). The gate insulating film **16** may be formed by thermal oxidation of a semiconductor crystal surface.

(72) Thereafter, a material for the gate electrode **17** and for the relay portion **21** (in the present preferred embodiment, polysilicon) is deposited on the n⁻ type base layer **12** while adding impurities (in the present preferred embodiment, n type impurities) (S6), and then a polysilicon layer deposited thereon is subjected to patterning (S7). Hence, the gate electrode **17** and the relay portion **21** (n type layer **22**) are simultaneously formed.

(73) Thereafter, p type ions are selectively implanted into the relay portion **21** (n type layer **22**) through a mask (S8). Hence, the p type region **23** is formed at the surface portion of the n type layer **22**.

(74) Thereafter, the interlayer insulating film **25** is formed so as to cover the gate electrode **17** and the relay portion **21** (S9), and the contact holes **26** to **28** are formed in the interlayer insulating film **25** by means of photolithography.

(75) Thereafter, the source electrode film **5** and the gate electrode film **6** are each formed on the interlayer insulating film **25** as a surface metal (S10).

(76) Thereafter, the surface insulating film **31** is formed so as to cover the source electrode film **5** and the gate electrode film **6** (S11), and the pad opening **33** is formed in the surface insulating film **31** by means of photolithography (S12).

(77) Thereafter, the drain electrode **18** is formed on the rear surface of the n⁺ type drain layer **11**, thus making it possible to obtain the semiconductor device **1** mentioned above.

(78) In the semiconductor device **1**, a reverse bias is applied to a pn junction portion (parasitic diode) between the p type body region **13** and the n⁻ type base layer **12** when a power source is connected between the source electrode film **5** and the drain electrode **18** under the condition that the drain electrode **18** is set as a high potential side whereas the source electrode film **5** is set as a low potential side. At this time, if a control voltage lower than a predetermined threshold voltage is applied to the gate electrode **17**, no current path is created in the drain-to-source space. In other words, the semiconductor device **1** reaches an OFF state. On the other hand, if a control voltage equal to or more than the threshold voltage is applied to the gate electrode **17**, electrons are attracted to a surface of the channel region **20**, and an inversion layer (channel) is formed. Hence, an electrically conductive state is reached between the n⁺ type source region **14** and the n⁻ type base layer **12**. In other words, a current path is created that reaches the drain electrode **18** from the source electrode film **5** through the n⁺ type source region **14**, the inversion layer of the channel region **20**, and the n⁻ type base layer **12** in this order. In other words, the semiconductor device **1**

reaches an ON state.

(79) In the thus-performed on-off operation, when a voltage is applied from the gate pad **32** to the gate electrode, ringing easily occurs when it is turned on, whereas ringing does not easily occur when it is turned off, and therefore it is preferable to reduce only the noise that is made when it is turned on.

(80) Therefore, in the semiconductor device **1**, the relay portion **21** is provided, and the first conductive film **9** (gate pad) is connected to both the p type region **23** and the n type layer **22** through the first contact **29**, and the second conductive film **10** (gate finger) is connected to only the p type region **23** through the second contact **30**.

(81) When a positive voltage with respect to the second conductive film **10** is applied to the first conductive film **9**, the flow of an electric current between the first conductive film **9** and the second conductive film **10** takes a direction from the first conductive film **9** toward the second conductive film **10**. In this case, a reverse current will flow through the diode **34**. Therefore, the current path is limited to the path of (1) the first conductive film **9**.fwdarw.the first contact **29**.fwdarw.the p type region **23**.fwdarw.the second contact **30**.fwdarw.the second conductive film **10**, and an electric current does not flow or hardly flows to the path of (2) the first conductive film **9**.fwdarw.the first contact **29**.fwdarw.the n type layer **22**.fwdarw.the diode **34**.fwdarw.the p type region **23**.fwdarw.the second contact **30**.fwdarw.the second conductive film **10** as shown in FIG. **4** (solid line arrows) and FIG. **8**.

(82) On the other hand, when a positive voltage with respect to the first conductive film **9** is applied to the second conductive film **10**, the flow of an electric current between the first conductive film **9** and the second conductive film **10** takes a direction from the second conductive film **10** toward the first conductive film **9**. In this case, a forward current will flow through the diode **34**. Therefore, it is possible to use two paths in total as current paths, i.e., it is possible to use the path of (3) the second conductive film **10**.fwdarw.the second contact **30**.fwdarw.the p type region **23**.fwdarw.the first contact **29**.fwdarw.the first conductive film **9** and the path of (4) the second conductive film **10**.fwdarw.the second contact **30**, the p type region **23**.fwdarw.the diode **34**.fwdarw.the n type layer **22**.fwdarw.the first contact **29**.fwdarw.the first conductive film **9** as shown in FIG. **4** (alternate long and short dash line arrows) and FIG. **9**.

(83) In other words, when it is turned on, the number of current paths is one, hence making it possible to relatively heighten resistance, and when it is turned off, the number of current paths is two, hence making it possible to relatively make resistance lower than that when turned on. As thus described, the number of paths of gate current that flows when the MISFET is turned on differs from the number of paths of gate current that flows when the MISFET is turned off, and the resistance when the MISFET is turned on differs from the resistance when the MISFET is turned off. Therefore, it is possible to appropriately control the behavior of a gate current when the MISFET is turned on/off. Moreover, it is possible to perform such current control inside the semiconductor device **1**, and therefore it is also possible to maintain the space efficiency when the semiconductor device **1** is mounted.

(84) Thereafter, in the structure of FIG. **4**, I-V characteristics of the gate electrode were examined by simulations when a positive voltage (30 V, pn.fwdarw.p) was applied to the first contact **29** and when a positive voltage (30 V, p.fwdarw.pn) was applied to the second contact **30** between the first contact **29** (pn) and the second contact **30** (p). FIG. **10** is a diagram showing I-V characteristics when it is turned on, and FIG. **11** is a diagram showing I-V characteristics when it is turned off. From a comparison between FIG. **10** and FIG. **11**, it has been understood that the rise of an electric current when it is turned on is smoother than that when turned off and that there is a great difference in electric resistance between when it is turned on and turned off.

(85) Although the preferred embodiment of the present invention has been described as above, the present invention can be embodied in other modes.

(86) For example, referring to FIG. **12**, the semiconductor device **1** may have a super-junction

structure including a p type column layer **35** formed at a portion below the p type body region **13**. In this case, the p type column layer **35** may be formed so as to be continuous with the p type body region **13** as shown in FIG. **12**, or may be disposed in such a manner as to be separated from the p type body region **13** as shown in FIG. **13**.

(87) Additionally, the second contact **30** is electrically connected to only the p type region **23** as described in the above preferred embodiment. The reason is that it is preferable to reduce only the noise caused when it is turned on because, with respect to a gate current, ringing easily occurs when it is turned on, whereas ringing does not easily occur when it is turned off. However, if there is an intention to relatively lower resistance when it is turned on and to relatively heighten resistance when it is turned off without being limited to the gate current, the second contact **30** may be electrically connected to only the n type layer **22** as shown in FIG. **14**.

(88) Additionally, although the relay portion **21** is disposed at a portion below the first and second conductive films **9** and **10** as described in the above preferred embodiment, the relay portion **21** may be formed so as to straddle the first conductive film **9** and the second conductive film **10** above the first and second conductive films **9** and **10**.

(89) Additionally, the relay portion **21** can also be made of, for example, aluminum, copper, or the like without being limited to polysilicon.

(90) Additionally, the structure of the unit cells **19** may be a planar gate structure as described in the above preferred embodiment, or may be a trench gate structure.

(91) Additionally, a configuration in which the conductivity type of each semiconductor part of the semiconductor device **1** is reversed may be employed. For example, in the semiconductor device **1**, the p type part may be an n type, and the n type part may be a p type.

(92) Besides, various design changes can be made within the scope of the matters described in the appended claims.

(93) This application corresponds to Japanese Patent Application No. 2018-64795 filed with the Japan Patent Office on Mar. 29, 2018, the entire disclosure of which is incorporated herein by reference.

EXAMPLES

(94) Next, the present invention will be described on the basis of examples, and yet the present invention is not restricted by the following examples.

Examples 1 to 7

(95) First, a measurement circuit of a semiconductor device of each of Examples 1 to 7 is as shown in FIG. **15**. In FIG. **15**, a part surrounded by the alternate long and two short dashes line A corresponds to the aforementioned semiconductor device **1**.

(96) Next, the structure of a relay portion **39** in semiconductor devices of Examples 1 to 7 will be concretely described with reference to FIG. **16** to FIG. **22**. FIG. **16** is Example 1 (pattern A), FIG. **17** is Example 2 (pattern B), FIG. **18** is Example 3 (pattern C), FIG. **19** is Example 4 (pattern D), FIG. **20** is Example 5 (pattern E), FIG. **21** is Example 6 (pattern F), and FIG. **22** is Example 7 (pattern G).

(97) Unlike the aforementioned relay portion **21**, the relay portion **39** is made of a polysilicon layer, which is common in the semiconductor devices of Examples 1 to 7. The relay portion **39** is composed of an n type layer **36** and a p type layer **37** that adjoins the n type layer **36** and that is contiguous to the n type layer **36**. Additionally, in the relay portion **39**, a slit **38** that partially separates the n type layer **36** and the p type layer **37** from each other is formed on an extension line of the boundary portion **24** between the n type layer **36** and the p type layer **37**.

(98) The relay portion **39** having the thus-formed structure can be manufactured as follows. First, a material (polysilicon) for the gate electrode **17** and a material (polysilicon) for the relay portion **39** are deposited on the n- type base layer **12** in the manner of step S6 of FIG. **7**. Thereafter, p type ions (boron) are implanted into the entire surface of the polysilicon layer to turn the polysilicon layer into a p type. Thereafter, the polysilicon layer is subjected to patterning in the manner of step

S7 of FIG. 7. Hence, the gate electrode **17** and the relay portion **39** are simultaneously formed. (99) Thereafter, p type ions (boron) are again implanted into the relay portion **39** when the p type body region **13** is formed. Thereafter, n type ions (arsenic) are implanted in a state in which the polysilicon layer is selectively covered with a mask (resist) when the n+ type source region **14** is formed. Hence, the n type layer **36** is formed at the relay portion **39**, and the p type layer **37** is also formed at the part covered with the mask. Thereafter, apart of the relay portion **39** is removed, and, as a result, the slit **38** is formed.

(100) Unlike the aforementioned preferred embodiment, the contact hole **27** is divided into a one-side contact hole **27A** by which the p type layer **37** is exposed and an opposite-side contact hole **27B** by which the n type layer **36** is exposed in the semiconductor devices of Examples 1 to 7. Additionally, the first contact **29** is divided into a one-side first contact **29A** connected to the p type layer **37** through the one-side contact hole **27A** and an opposite-side first contact **29B** connected to the n type layer **36** through the opposite-side contact hole **27B**. In other words, the one-side first contact **29A** and the opposite-side first contact **29B** are formed so as to be independent of each other.

(101) The semiconductor devices of Examples 1 to 7 that have been structured as above were incorporated into the circuit of FIG. **15**, and time-dependent changes in electric current caused when the gate voltage is turned on/off were verified. Results are shown in FIGS. **23** to **25**. The term "Ref" shown in FIG. **23** to FIG. **25** denotes a structure in which the first conductive film **9** and the second conductive film **10** are short-circuited without providing the relay portion **39**. Additionally, the pattern is changed on the basis of Example 2 in the following description.

(102) First, Examples 1 to 3 are compared with each other in FIG. **23**. A difference among pattern A, pattern B, and pattern C is the size of the area of the p type layer **37** as shown in FIG. **16** to FIG. **18** (the position of the one-side first contact **29A** and the position of the opposite-side first contact **29B** are fixed). Hence, resistance values of the p type layer **37** differ from each other between the one-side first contact **29A** and the second contact **30**.

(103) It was confirmed how the drain current changes in accordance with the difference in area of the p type layer **37** when the gate voltage is turned on/off. As a result, when the gate voltage was turned on, ringing (near $-0.7\ \mu\text{s}$, near $0.5\ \mu\text{s}$) was suppressed compared with Ref in all of Examples 1 to 3. On the other hand, when the gate voltage was turned off, the discharging rate was substantially equal to that of Ref because the number of current paths becomes two as shown by the alternate long and short dash line arrows of FIG. **4**.

(104) Next, Examples 2, 4, and 5 are compared with each other in FIG. **24**. A difference among pattern B, pattern D, and pattern E is the position of the opposite-side first contact **29B** as shown in FIG. **17**, FIG. **19**, and FIG. **20** (the size of the area of the p type layer **37** and the position of the one-side first contact **29A** are fixed). Hence, resistance values of the n type layer **36** differ from each other between the opposite-side first contact **29B** and the second contact **30**.

(105) It was confirmed how the drain current changes in accordance with the difference in position of the opposite-side first contact **29B** when the gate voltage is turned on/off. As a result, when the gate voltage was turned on, ringing (near $-0.7\ \mu\text{s}$, near $0.5\ \mu\text{s}$) was suppressed compared with Ref in both of Examples 4 and 5. On the other hand, when the gate voltage was turned off, the discharging rate was substantially equal to that of Ref because the number of current paths becomes two as shown by the alternate long and short dash line arrows of FIG. **4**.

(106) Next, Examples 2, 6, and 7 are compared with each other in FIG. **25**. A difference among pattern B, pattern F, and pattern G is the position of the one-side first contact **29A** as shown in FIG. **17**, FIG. **21**, and FIG. **22** (the size of the area of the p type layer **37** and the position of the opposite-side first contact **29B** are fixed). Hence, resistance values of the p type layer **37** differ from each other between the one-side first contact **29A** and the second contact **30**.

(107) It was confirmed how the drain current changes in accordance with the difference in position of the one-side first contact **29A** when the gate voltage is turned on/off. As a result, when the gate

voltage was turned on, ringing (near $-0.7\ \mu\text{s}$, near $0.5\ \mu\text{s}$) was suppressed compared with Ref in both of Examples 6 and 7. On the other hand, when the gate voltage was turned off, the discharging rate was substantially equal to that of Ref because the number of current paths becomes two as shown by the alternate long and short dash line arrows of FIG. 4.

(108) As described above, ringing was suppressed compared with Ref in all of Examples 1 to 7 when the gate voltage was turned on.

(109) On the other hand, a great change in characteristics was found in accordance with a change in position of the one-side first contact **29A** as shown in FIG. 25. In other words, the discharging rate became slower in proportion to an approach of the one-side first contact **29A** to the second contact **30**. On the other hand, a great change in characteristics was not produced with a change in area size of the p type layer **37** and with a change in position of the opposite-side first contact **29B**.

Therefore, it has been understood that it is recommended to change the position of the one-side first contact **29A** if there is an intention to greatly change characteristics, and it is recommended to change the size of the area of the p type layer **37** or change the position of the opposite-side first contact **29B** if there is an intention to finely adjust characteristics.

Examples 8 and 9

(110) Next, structures of relay portions **40** and **41** in semiconductor devices of Examples 8 and 9 will be concretely described with reference to FIG. 26 and FIG. 27. FIG. 26 is Example 8 (pattern H), and FIG. 27 is Example 9 (pattern I).

(111) In Example 8, the relay portion **40** is composed of two polysilicon layers **42** and **43**. The polysilicon layer **42**, which is one of the two polysilicon layers, is formed as a p type layer in its entirety. The polysilicon layer **42** is connected to the one-side first contact **29A** and to the one-side second contact **30A**. The polysilicon layer **43**, which is the other one of the two polysilicon layers, is partitioned into an n type layer **44** and a p type layer **45**. The p type layer **45** is connected to the opposite-side second contact **30B**, and the n type layer **44** is connected to the opposite-side first contact **29B**.

(112) In Example 9, the relay portion **41** includes an n type layer **46** and a p type layer **47**. The n type layer **46** straddles the first conductive film **9** and the second conductive film **10** below the first conductive film **9** and the second conductive film **10**. The p type layer **47** surrounds the n type layer **46**, and is contiguous to the n type layer **46**. The p type layer **47** is connected to the one-side first contact **29A** and to the second contact **30**. The n type layer **46** is connected to the opposite-side first contact **29B**.

(113) Likewise, in Examples 8 and 9, verification was conducted by use of the aforementioned measurement circuit of FIG. 15, and, as a result, ringing was suppressed compared with Ref when the gate voltage was turned on in the same way as in Examples 1 to 7 (not shown).

REFERENCE SIGNS LIST

(114) **1** Semiconductor device **2** Semiconductor substrate **3** Active region **4** Outer peripheral portion **6** Gate electrode film **9** First conductive film **10** Second conductive film **11** N+ type drain layer **12** N- type base layer **13** P type body region **14** N+ type source region **16** Gate insulating film **17** Gate electrode **19** Unit cell **21** Relay portion **22** N type layer **23** P type region **24** Boundary portion **29** First contact **29A** One-side first contact **29B** Opposite-side first contact **30** Second contact **31** Surface insulating film **34** Diode **36** N type layer **37** P type layer **38** Slit **39** Relay portion **40** Relay portion **41** Relay portion **42** Polysilicon layer **43** Polysilicon layer **44** N type layer **45** P type layer **46** N type layer **47** P type layer

Claims

1. A semiconductor device comprising: a semiconductor layer; a first conductor disposed on the semiconductor layer; a second conductor disposed on the semiconductor layer so as to be separated from the first conductor; a relay portion that is formed on the semiconductor layer so as to straddle

the first conductor and the second conductor and that is made of a semiconductor having a first conductivity type region and a second conductivity type region; a first contact by which the relay portion is electrically connected to the first conductor; and a second contact that electrically connects the relay portion and the second conductor together, wherein a first current path formed in the relay portion when a positive voltage with respect to the first conductor is applied to the second conductor has relatively lower resistance than a second current path formed in the relay portion when a positive voltage with respect to the second conductor is applied to the first conductor, and the relay portion is disposed closer to the semiconductor layer than a gate pad and a gate wiring.

2. The semiconductor device according to claim 1, further comprising a functional element formed at the semiconductor layer, wherein the first conductor includes an external terminal to which electric power is supplied from outside, and the second conductor includes a wiring that supplies electric power supplied to the first conductor to the functional element.

3. The semiconductor device according to claim 2, wherein the functional element is an element including a gate electrode that controls an electric current that flows to the functional element, and the external terminal includes the gate pad, to which an electroconductive bonding member is bonded from outside, and the wiring includes the gate wiring, that supplies electric power supplied to the gate pad to the gate electrode, the first conductivity type region is a p type region, and the second conductivity type region is an n type region.

4. The semiconductor device according to claim 3, wherein the gate wiring includes a gate finger disposed at an outer peripheral portion of the semiconductor layer so as to surround the functional element.

5. The semiconductor device according to claim 3, wherein the functional element includes a field-effect transistor that has a body region selectively formed at a surface portion of the semiconductor layer, a source region formed at an inner portion of the body region, and the gate electrode facing a part of the body region through a gate insulating film.

6. The semiconductor device according to claim 5, wherein the body region includes a plurality of body regions that extend in a striped manner with intervals from each other.

7. The semiconductor device according to claim 1, wherein the relay portion includes a second conductivity type layer as the second conductivity type region and the first conductivity type region selectively formed at a surface portion of the second conductivity type layer.

8. The semiconductor device according to claim 7, wherein a thickness of the second conductivity type layer is $0.1\ \mu\text{m}$ to $10\ \mu\text{m}$, and a depth of the first conductivity type region from a surface of the second conductivity type layer is $0.1\ \mu\text{m}$ to $10\ \mu\text{m}$.

9. The semiconductor device according to claim 1, wherein the relay portion includes a second conductivity type layer as the second conductivity type region and a first conductivity type layer as the first conductivity type region, the first conductivity type layer adjoining the second conductivity type layer and being contiguous to the second conductivity type layer.

10. The semiconductor device according to claim 9, further comprising a slit that is formed on an extension line of a boundary portion between the second conductivity type layer and the first conductivity type layer and by which the second conductivity type layer and the first conductivity type layer are partially separated from each other.

11. The semiconductor device according to claim 1, wherein the first contact is formed so as to straddle the first conductivity type region and the second conductivity type region of the relay portion.

12. The semiconductor device according to claim 1, wherein the first contact includes a one-side first contact that is connected to the first conductivity type region of the relay portion and an opposite-side first contact that is apart from the one-side first contact and that is connected to the second conductivity type region of the relay portion.

13. The semiconductor device according to claim 1, wherein an impurity concentration of the first conductivity type region is $1.0 \times 10^{19}\ \text{cm}^{-3}$ to $1.0 \times 10^{21}\ \text{cm}^{-3}$, and an impurity

concentration of the second conductivity type region is 1.0×10^{19} cm.^{sup.}−3 to 1.0×10^{21} cm.^{sup.}−3.

14. The semiconductor device according to claim 1, wherein the first conductor and the second conductor are each made of aluminum, and the relay portion is made of polysilicon.

15. The semiconductor device according to claim 1, wherein the semiconductor layer includes a silicon substrate.

16. The semiconductor device according to claim 1, wherein the second conductor is formed in a linear shape along a side surface of the semiconductor layer.

17. The semiconductor device according to claim 1, wherein one of sides of the first conductor is disposed along a side surface of the semiconductor layer and faces the second conductor.
